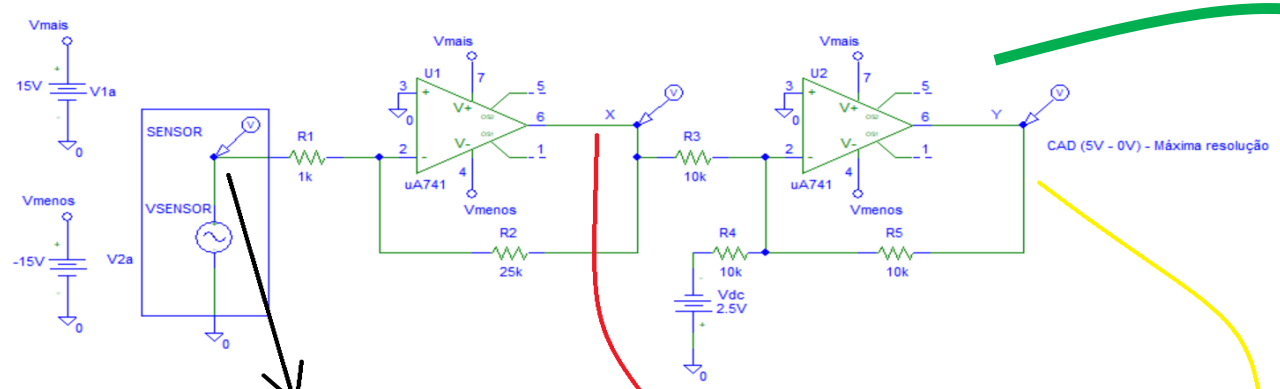
# CDA *versus* CAD Eletrónica

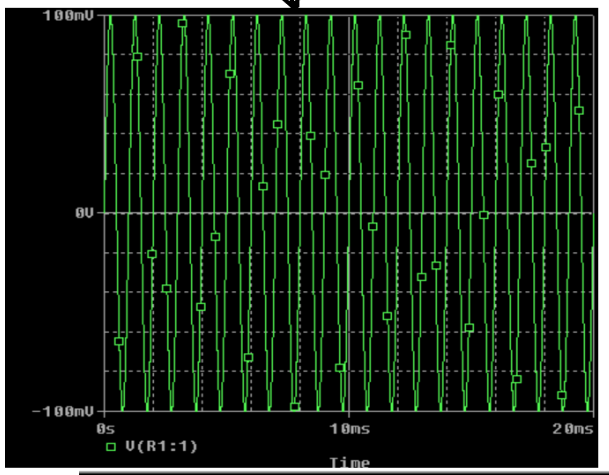
©Acácio Amaral

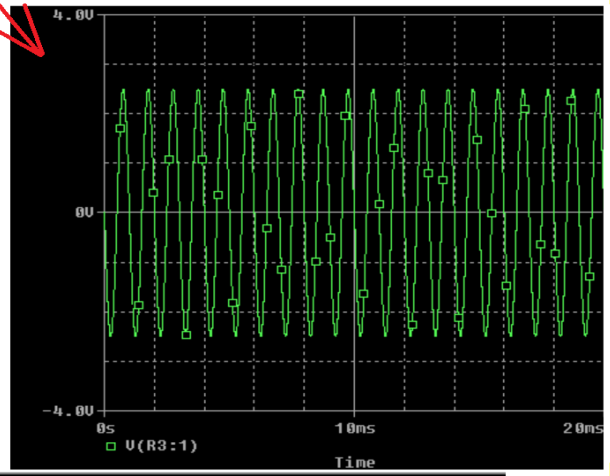
(não autorizo a republicação deste material)

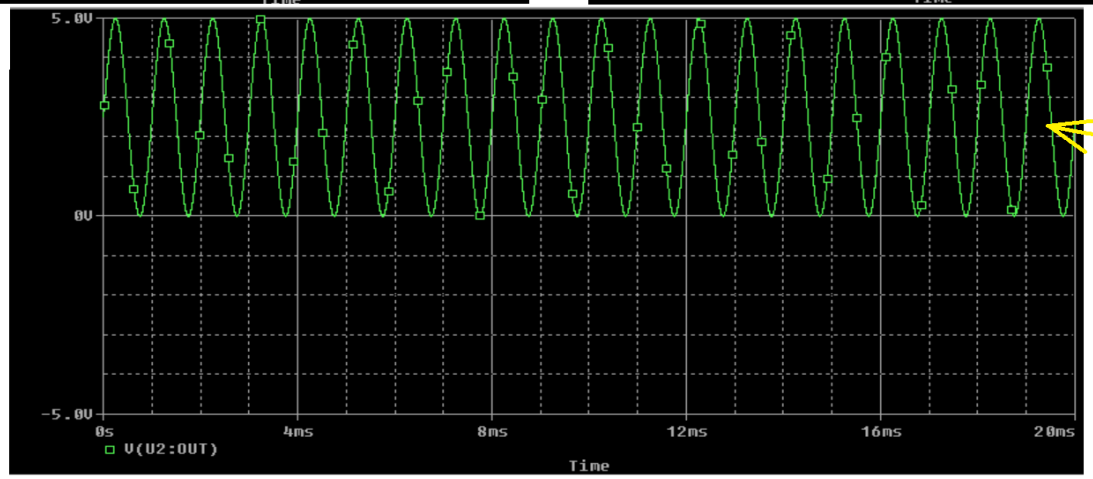
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05/01/2023




[1] →

## *Adaptar níveis de tensão*

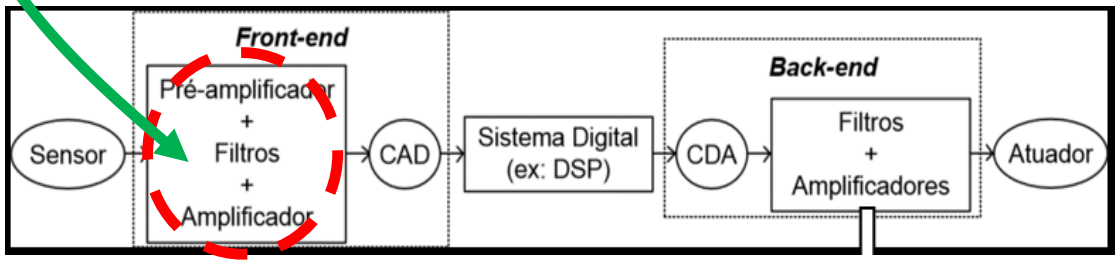
## *Sensor ( ? V) → Circuito acondicionamento de sinal → CAD (5V)*

## *Resolução = 0.625 V (3bits)*

## *{0, 0.625, 1.25, 1.875, 2.5, 3.125, 3.75, 4.375}*

## Computador analógico

## Circuito de acondicionamento de sinal




## *Processamento de Sinal Analógico*

## *Deslocamento e Escalonamento de Sinais de Analógicos*

$$Sinal_{saida} = V_{SENSOR} \times 25 + V_{dc}$$



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# Conversão Digital-Analógica

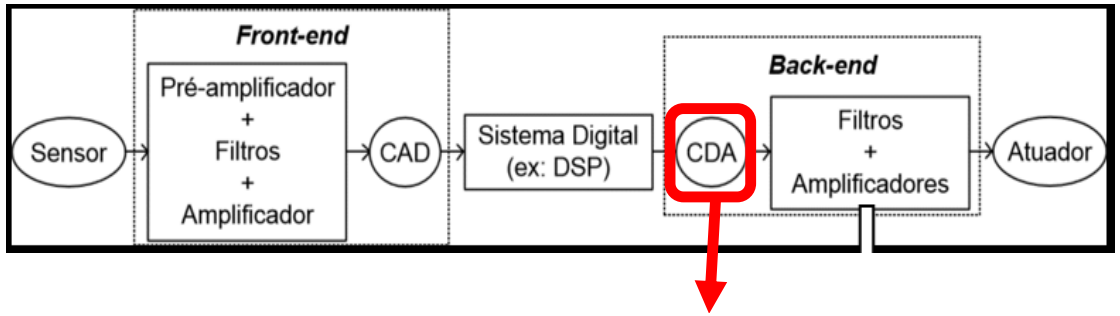
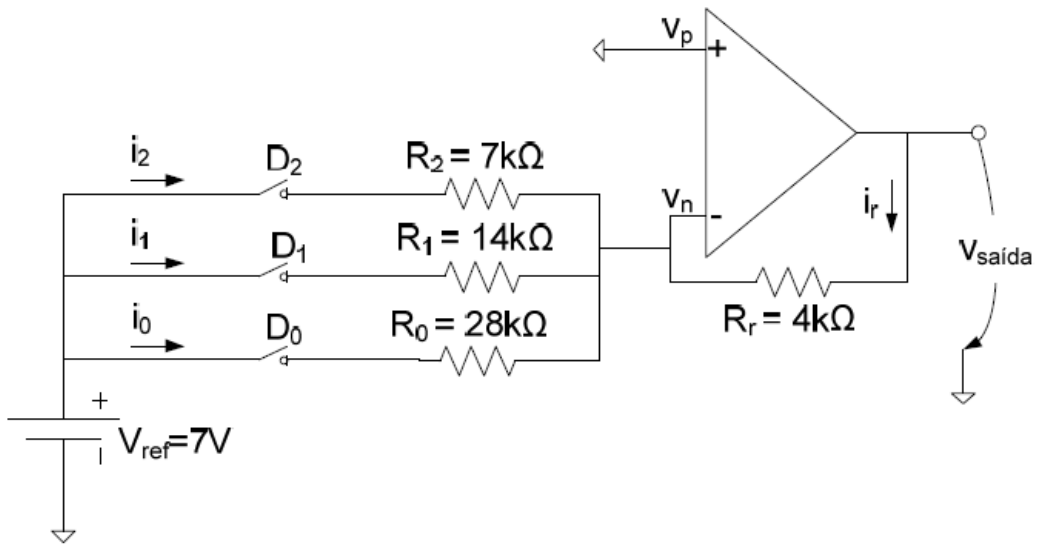



**Figura 6.27. CDA de 3 *bits* e resolução de 1 Volt**



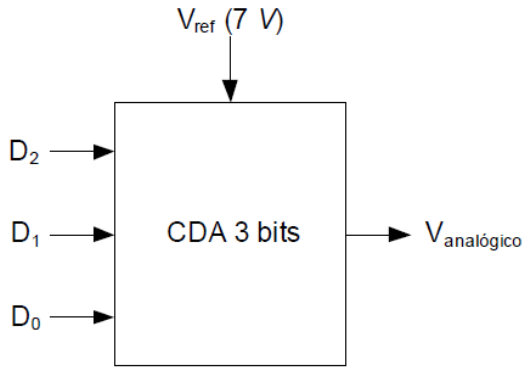

$$V_{saída} = -(4 \times D_2 + 2 \times D_1 + 1 \times D_0) \quad (6.47)$$

←[1]

## CDA → Montagem baseada na soma de correntes

**Figura 6.25. Conversor digital-analógico de 3 *bits* com resolução[1] de 1 Volt: (*a*) diagrama de blocos e (*b*) tabela funcional**

←[1] →



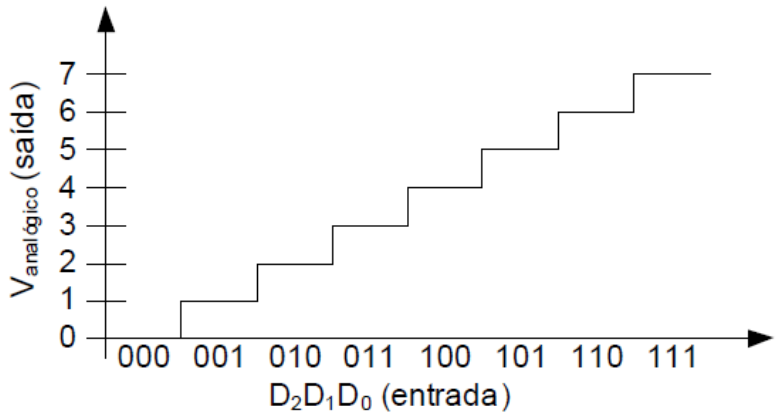

(a)

| $D_2$ | $D_1$ | $D_0$ | $V_{analógico}$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 2 |
| 0 | 1 | 1 | 3 |
| 1 | 0 | 0 | 4 |
| 1 | 0 | 1 | 5 |
| 1 | 1 | 0 | 6 |
| 1 | 1 | 1 | 7 |

(b)

**Figura 6.26. Função transferência do circuito da figura anterior**

←[1] →

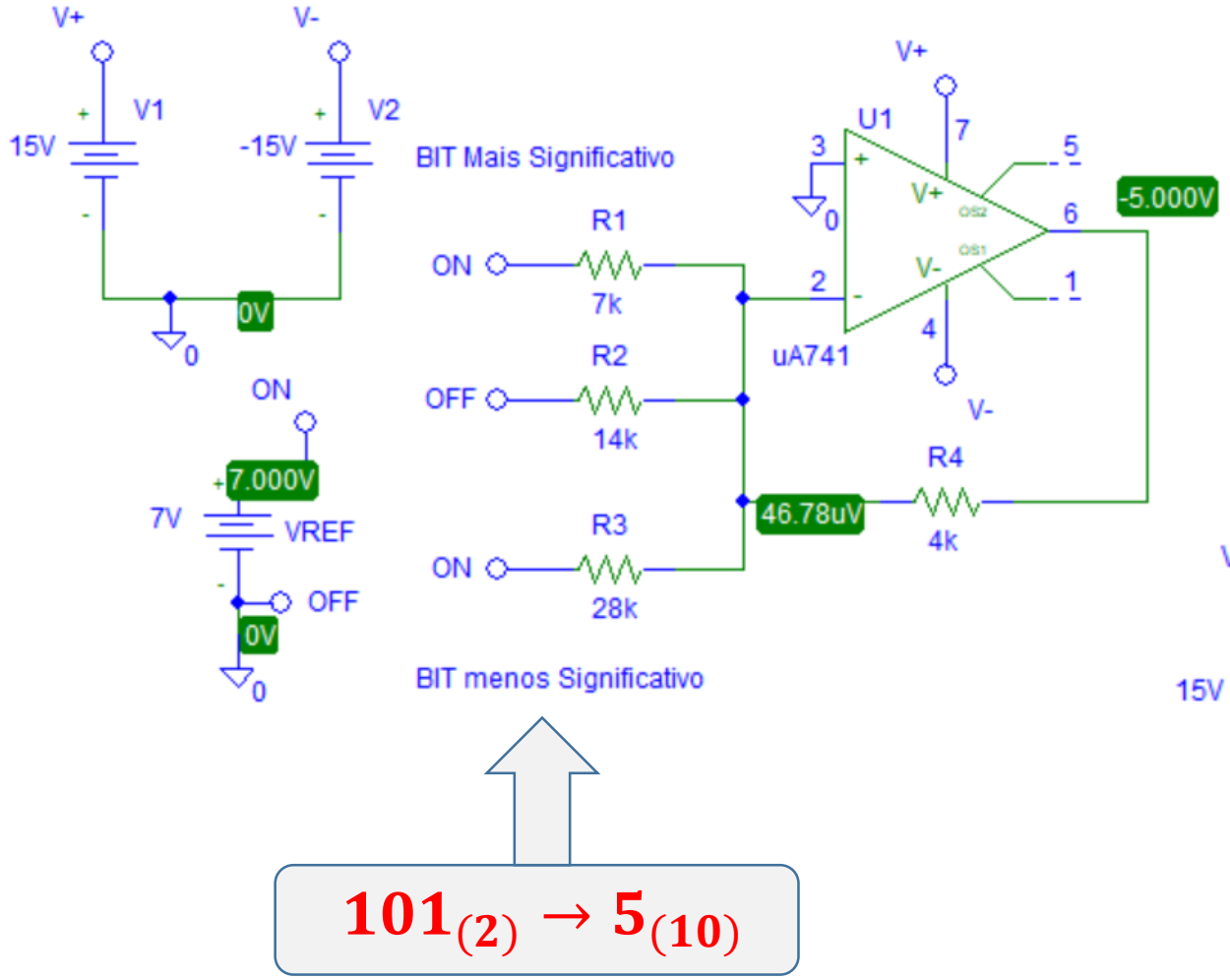




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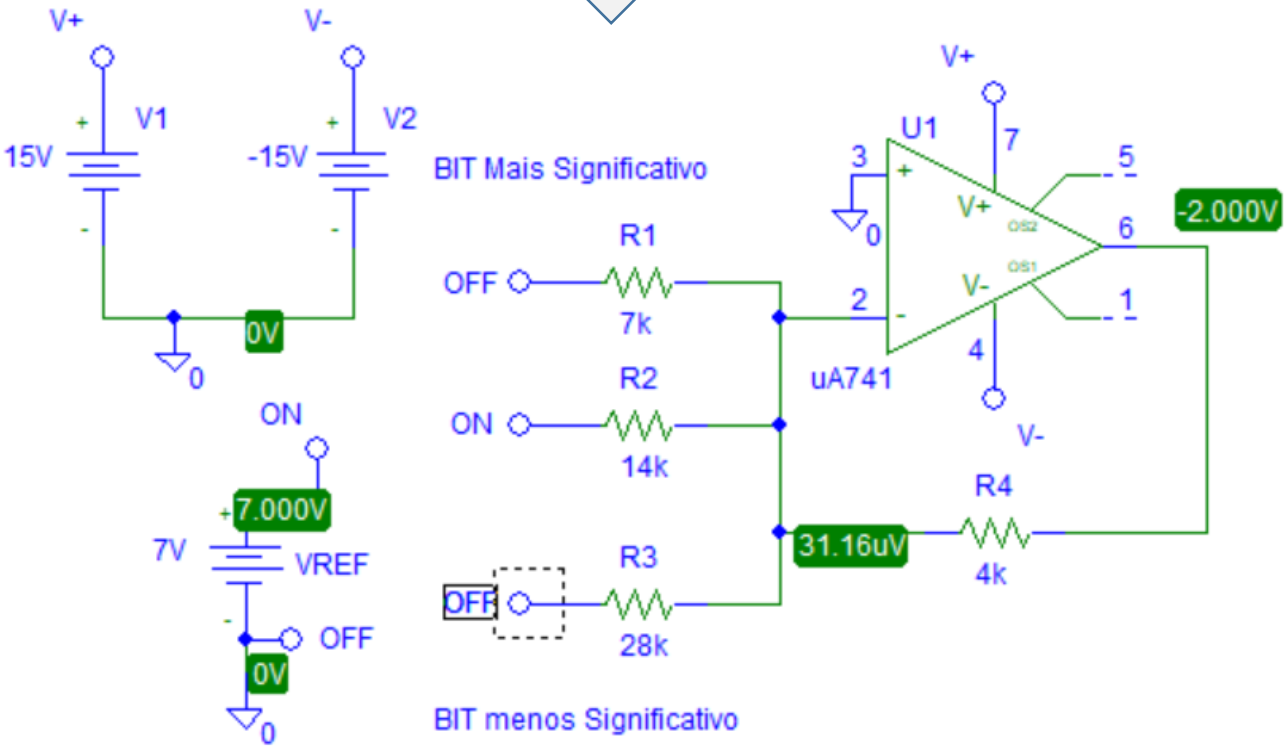
# CDA → Montagem baseada na soma de correntes

## *Conversão Digital-Analógica*




$101_{(2)} \rightarrow 5_{(10)}$

$010_{(2)} \rightarrow 2_{(10)}$



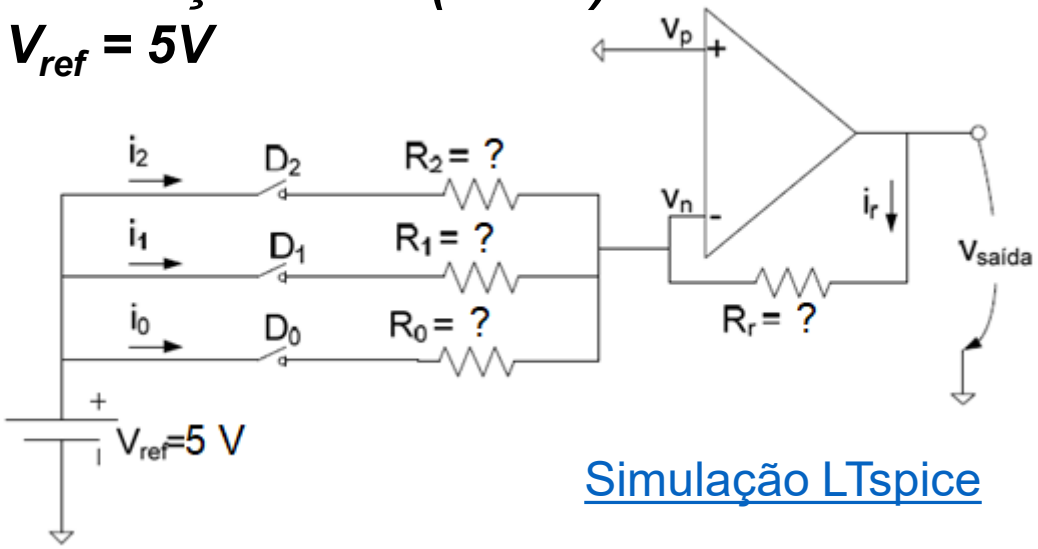

[Simulação Pspice]


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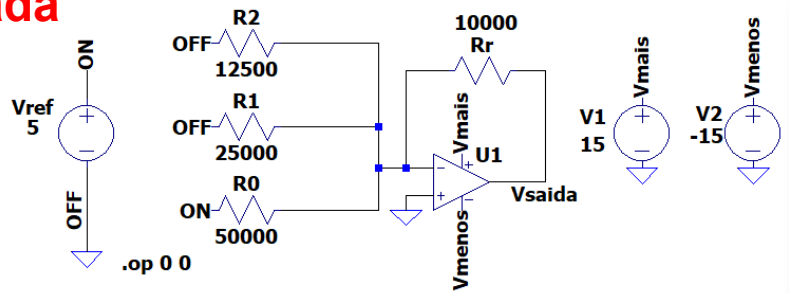
# *Projeto de um CDA:*

- ***Resolução = 1 V (3 Bits).***
- ***$V_{ref}$ = 5V***

**CDA → Montagem baseada na soma de correntes**



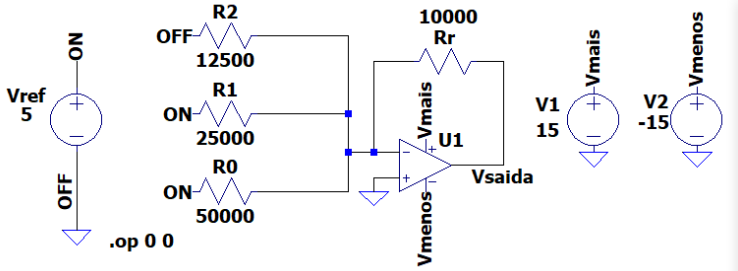

Simulação LTspice




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--- Operating Point ---

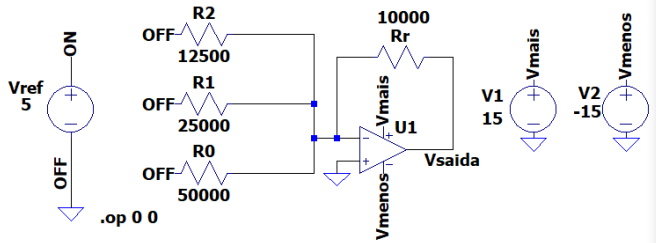
| | | |
|---|---|---|
| V(n001): | 1.00001e-006 | voltage |
| V(vsaida): | -0.999998 | voltage |
| V(vmais): | 15 | voltage |
| V(vmenos): | -15 | voltage |
| V(on): | 5 | voltage |
| I(Rr): | -9.99999e-005 | device_cu |
| I(R0): | -0.0001 | device_cu |
| I(R1): | 4.00003e-011 | device_cu |
| I(R2): | 8.00006e-011 | device_cu |




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--- Operating Point ---

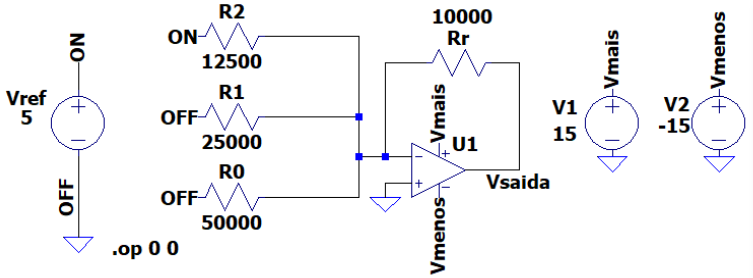
| | | |
|---|---|---|
| V(n001): | 3.00002e-006 | voltage |
| V(vsaida): | -2.99999 | voltage |
| V(vmais): | 15 | voltage |
| V(vmenos): | -15 | voltage |
| V(on): | 5 | voltage |
| I(Rr): | -0.0003 | device_cu |
| I(R0): | -9.99999e-005 | device_cu |
| I(R1): | -0.0002 | device_cu |
| I(R2): | 2.40002e-010 | device_cu |




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--- Operating Point ---

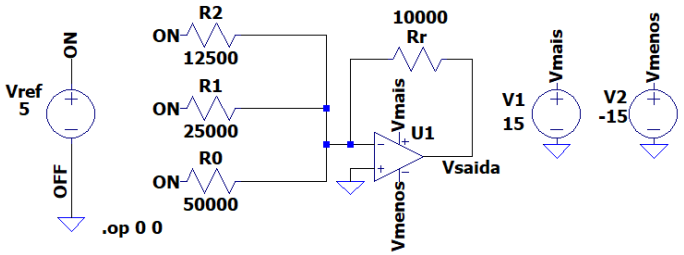
| | | |
|---|---|---|
| V(n001): | 0 | voltage |
| V(vsaida): | 0 | voltage |
| V(vmais): | 15 | voltage |
| V(vmenos): | -15 | voltage |
| V(on): | 5 | voltage |
| I(Rr): | 0 | device_current |
| I(R0): | 0 | device_current |
| I(R1): | 0 | device_current |
| I(R2): | 0 | device_current |
| I(V2): | 3e-008 | device_current |




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--- Operating Point ---

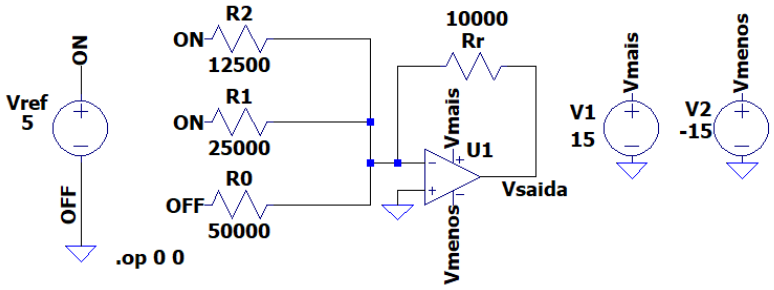
| | | |
|---|---|---|
| V(n001): | 4.00003e-006 | voltage |
| V(vsaida): | -3.99999 | voltage |
| V(vmais): | 15 | voltage |
| V(vmenos): | -15 | voltage |
| V(on): | 5 | voltage |
| I(Rr): | -0.000399999 | device_current |
| I(R0): | 8.00006e-011 | device_current |
| I(R1): | 1.60001e-010 | device_current |
| I(R2): | -0.0004 | device_current |




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--- Operating Point ---

| | | |
|---|---|---|
| V(n001): | 7.00005e-006 | voltage |
| V(vsaida): | -6.99998 | voltage |
| V(vmais): | 15 | voltage |
| V(vmenos): | -15 | voltage |
| V(on): | 5 | voltage |
| I(Rr): | -0.000699999 | device_cu |
| I(R0): | -9.99999e-005 | device_cu |
| I(R1): | -0.0002 | device_cu |
| I(R2): | -0.000399999 | device_cu |
| I(V2): | 3e-008 | device_cu |
| I(V1): | -3e-008 | device_cu |



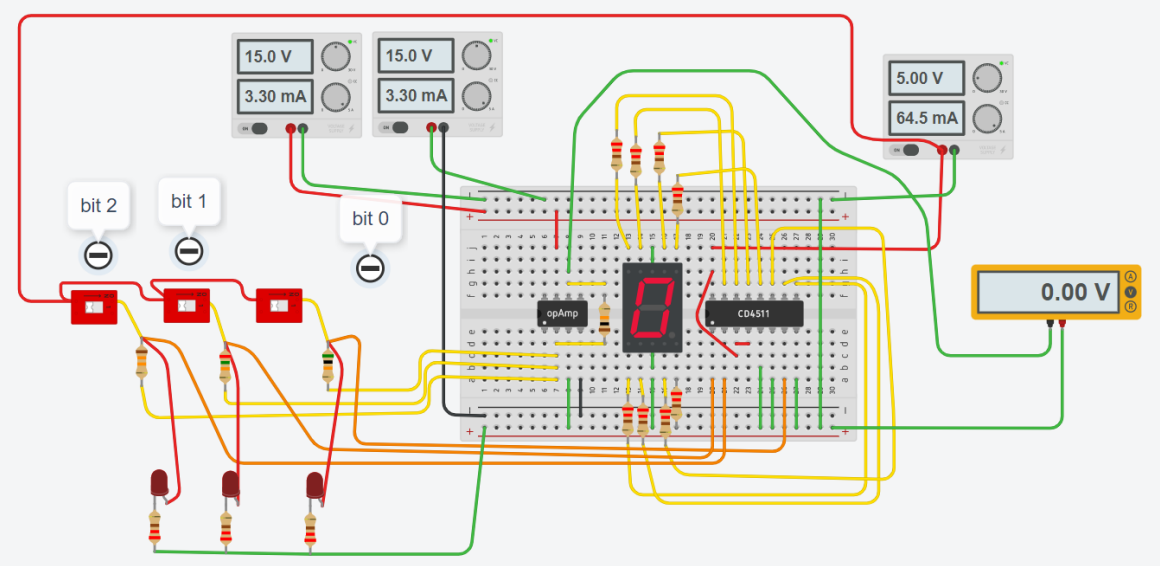

* C:\ACACIO\ISEC\Lectiva\2022_2023\eletronica\Aulas Teoricas\

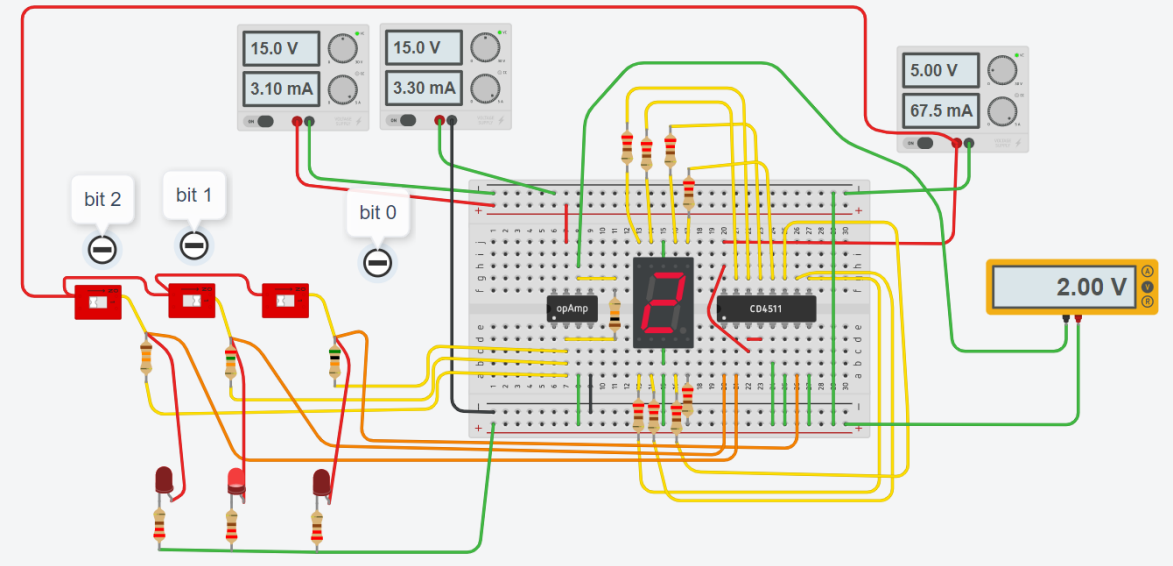
--- Operating Point ---

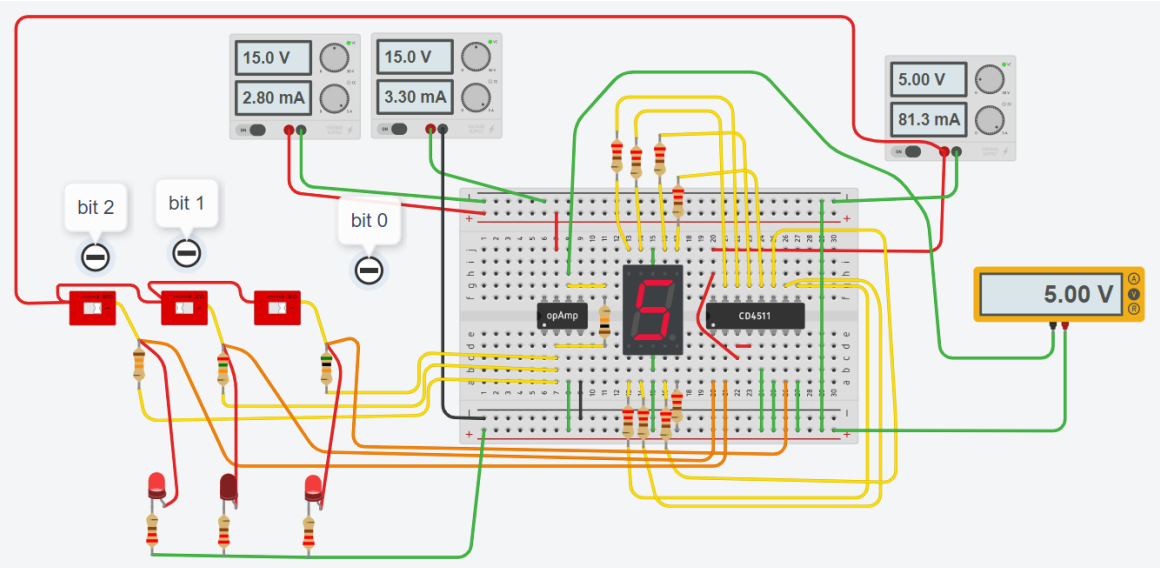
| | | |
|---|---|---|
| V(n001): | 6.00005e-006 | voltage |
| V(vsaida): | -5.99999 | voltage |
| V(vmais): | 15 | voltage |
| V(vmenos): | -15 | voltage |
| V(on): | 5 | voltage |
| I(Rr): | -0.000599999 | device_current |
| I(R0): | 1.20001e-010 | device_current |
| I(R1): | -0.0002 | device_current |

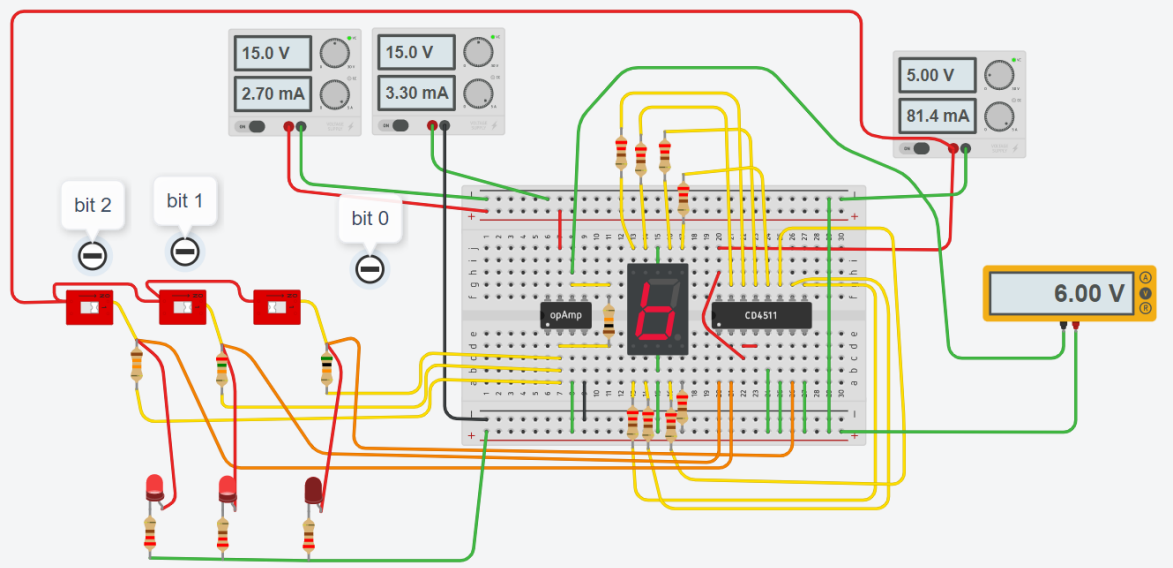


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**Conversão Digital-Analógica**

https://www.tinkercad.com/things/l85OYUSmAyg?sharecode=rCByNV7cjQVpAch3k7r973WQdIxsk4FsdCgQiRu9cM0

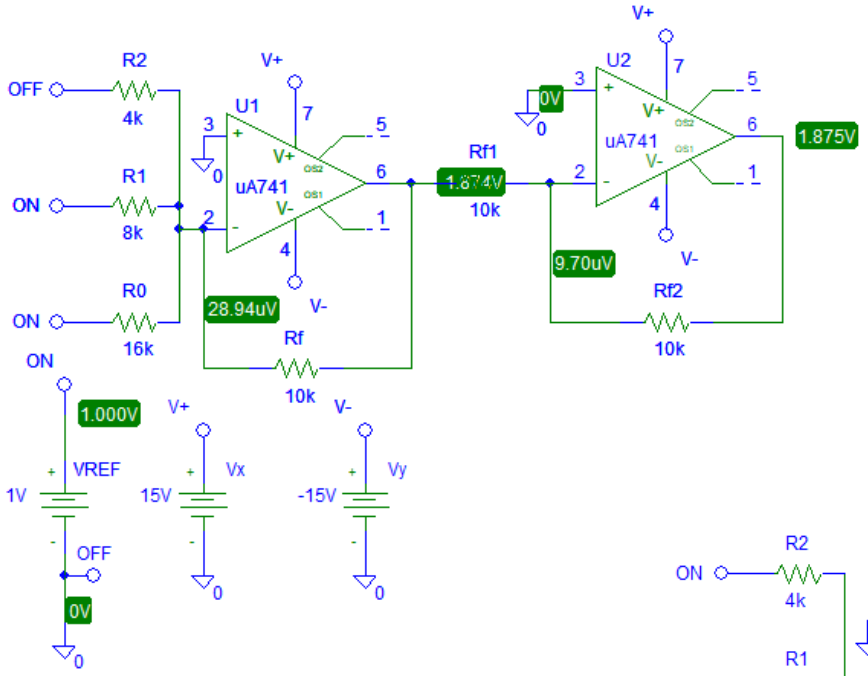
**CDA → Montagem baseada na soma de correntes**


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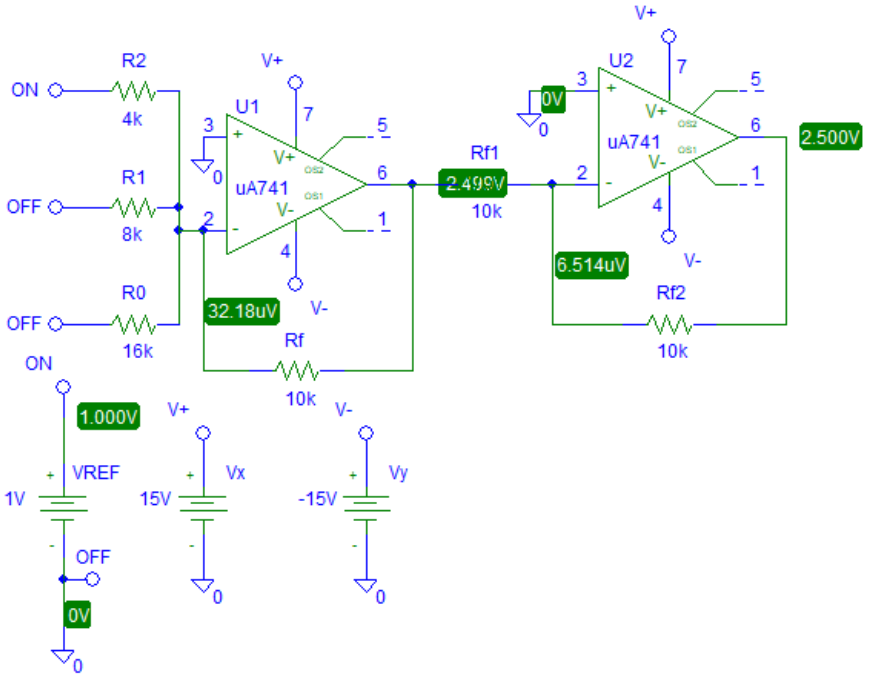
# Projeto: Resolução = 0.625 V (3bits)
# {0, 0.625, 1.25, 1.875, 2.5, 3.125, 3.75, 4.375}

[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

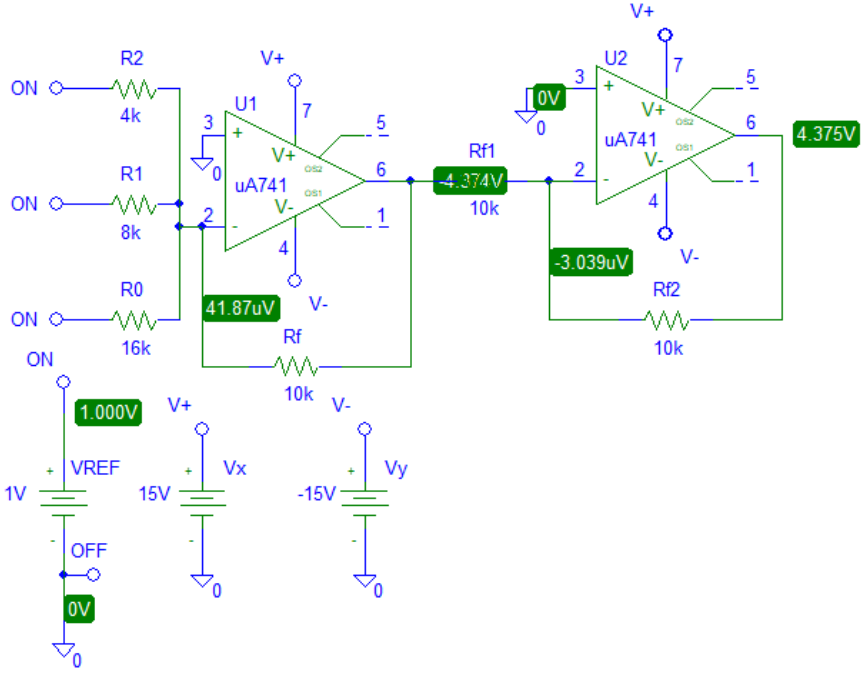
0 1 1




1 0 0




1 1 1





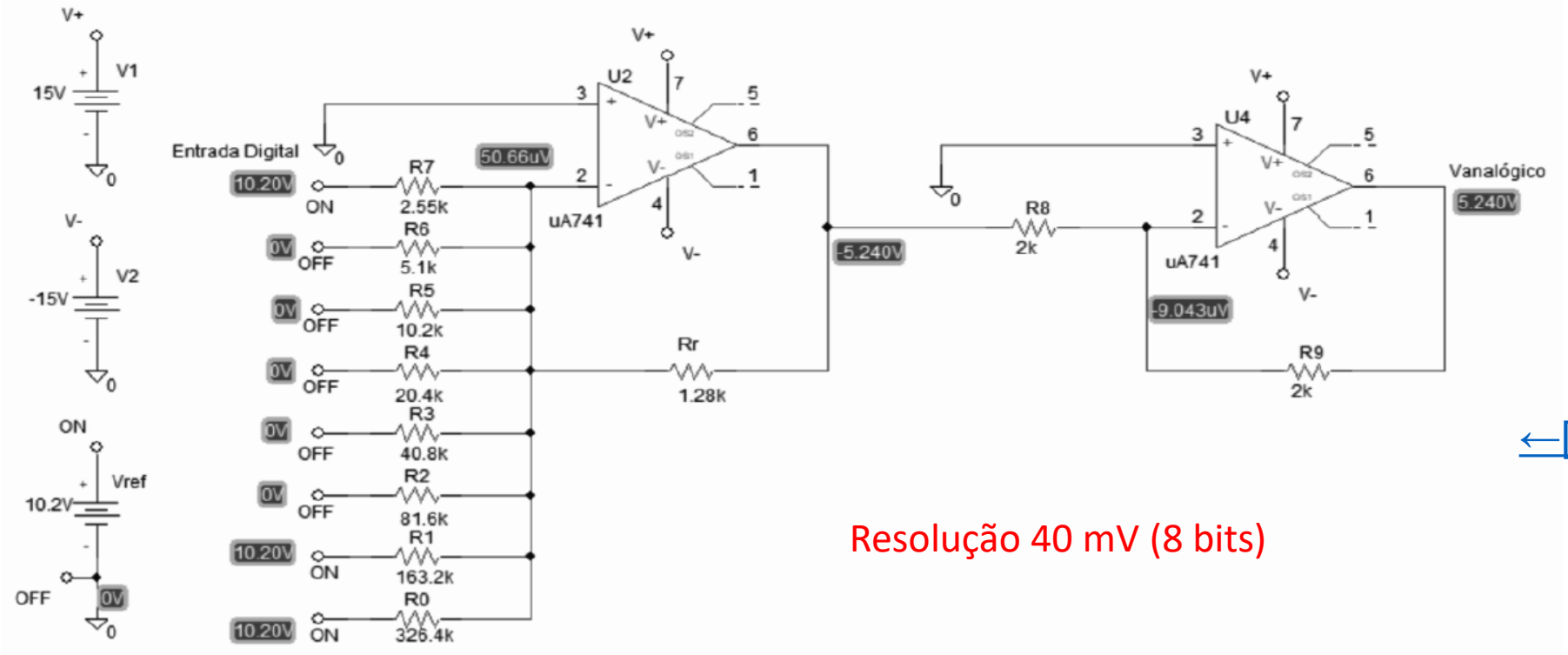
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Conversão Digital-Analógica

## Limitações:

- **CDA → Montagem baseada na soma de correntes :**
  - **Enorme gama de valores de resistência.**
  - **Nº total de bits limitado pela resistência de entrada do Amp-op (2 MΩ).**
  - **Tolerância das resistências (componentes reais).**

**Figura 6.28. Conversor digital-analógico projetado na secção anterior e implementado em Pspice**



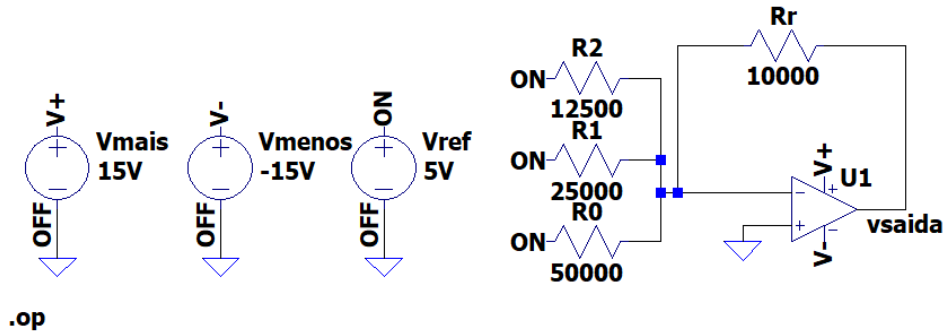

Resolução 40 mV (8 bits)

←[1]


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# Conversão Digital-Analógica

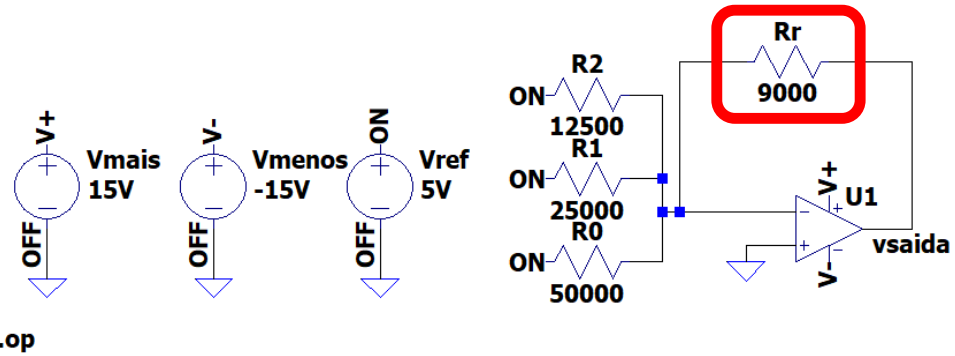
Exemplos → erros (tolerâncias resistências)




```
* C:\ACACIO\ISEC\Lectiva\2022_2023\eletronica\Aulas Teoricas\A
        --- Operating Point ---

V(n001):        7.00005e-006    voltage
V(vsaida):      -6.99998        voltage
V(v+):          15              voltage
V(v-):          -15             voltage
V(on):          5               voltage
I(Rr):          -0.000699999    device_current
I(R0):          -9.99999e-005   device_current
I(R1):          -0.0002         device_current
I(R2):          -0.000399999    device_current
I(Vmais):       -3e-008         device_current
I(Vmenos):      3e-008          device_current
```

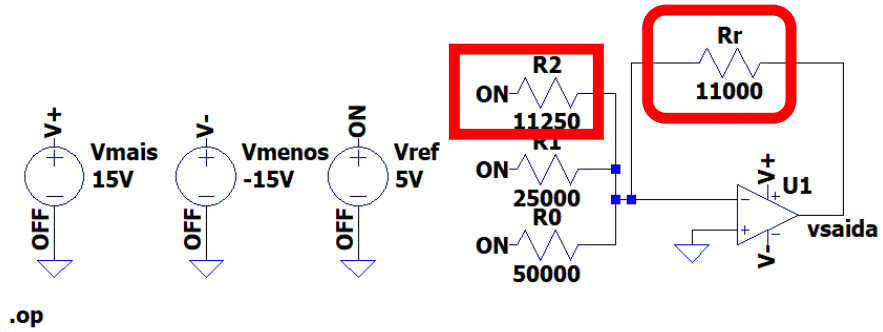
-7V




```
* C:\ACACIO\ISEC\Lectiva\2022_2023\eletronica\Aulas Teoricas\
        --- Operating Point ---

V(n001):        6.30006e-006    voltage
V(vsaida):      -6.29999        voltage
V(v+):          15              voltage
V(v-):          -15             voltage
V(on):          5               voltage
I(Rr):          -0.000699999    device_current
I(R0):          -9.99999e-005   device_current
I(R1):          -0.0002         device_current
I(R2):          -0.000399999    device_current
I(Vmais):       -3e-008         device_current
I(Vmenos):      3e-008          device_current
```

-6.3V




```
* C:\ACACIO\ISEC\Lectiva\2022_2023\eletronica\Aulas Teoricas\A
        --- Operating Point ---

V(n001):        8.18894e-006    voltage
V(vsaida):      -8.18887        voltage
V(v+):          15              voltage
V(v-):          -15             voltage
V(on):          5               voltage
I(Rr):          -0.000744443    device_current
I(R0):          -9.99998e-005   device_current
I(R1):          -0.0002         device_current
I(R2):          -0.000444444    device_current
I(Vmais):       -3e-008         device_current
I(Vmenos):      3e-008          device_current
I(Vref):        -0.000744443    device_current
```

-8.2V


[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

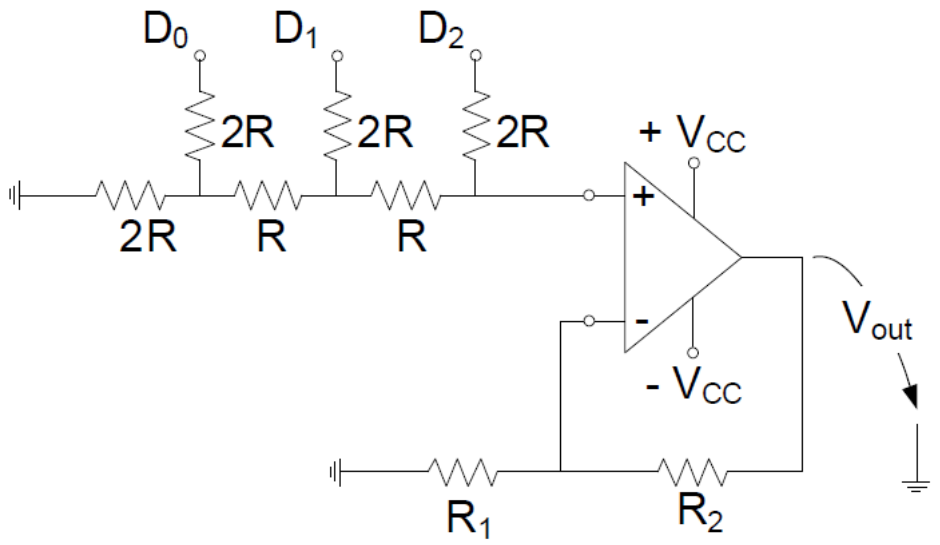




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# Conversão Digital-Analógica

**Figura 7.70. Conversor digital-analógico em escada R-2R (3 *bits*)**




←[1]

*Cálculos das resistências equivalentes (auxilia no cálculo final):*

$$Paralelo \rightarrow (2R\ com\ 3R) = \frac{12}{10} \times R$$

$$Paralelo \rightarrow (2R\ com\ 2.2R) = \frac{4.4}{4.2} \times R$$

$$v_{out} = \left( \frac{V_{ref} \times D_2}{2^1} + \frac{V_{ref} \times D_1}{2^2} + \frac{V_{ref} \times D_0}{2^3} \right) \times \left( \frac{R_2}{R_1} + 1 \right) \quad (7.71)$$

←[1]

$$v_{out} = V_{ref} \times \left( \frac{R_2}{R_1} + 1 \right) \times \sum_{i=0}^{N-1} \left( \frac{D_i}{2^{N-i}} \right) \quad (7.72)$$

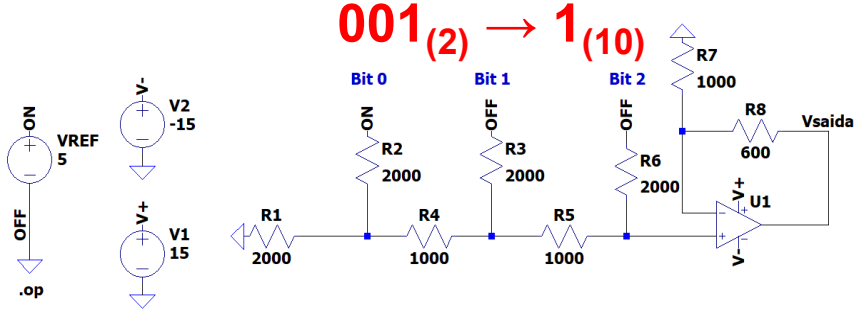
←[1]


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# CDA → Montagem baseada Escada R-2R

**Conversão Digital-Analógica**

## $001_{(2)} \rightarrow 1_{(10)}$

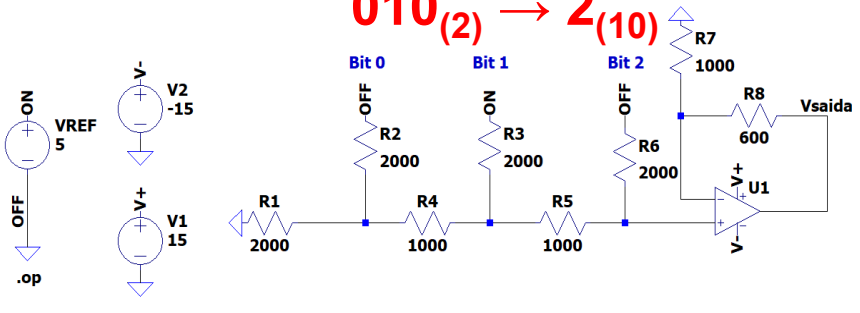



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--- Operating Point ---

| | | |
|---|---|---|
| V(n001) : | 0.624998 | voltage |
| V(n004) : | 0.624999 | voltage |
| V(vsaida) : | 0.999997 | voltage |
| V(v+) : | 15 | voltage |
| V(v-) : | -15 | voltage |
| V(n002) : | 1.71875 | voltage |
| V(on) : | 5 | voltage |
| V(n003) : | 0.937499 | voltage |
| I(R8) : | 0.000624999 | device_cu |
| I(R7) : | -0.000624998 | device_cu |
| I(R6) : | -0.000312499 | device_cu |
| I(R5) : | -0.000312501 | device_cu |
| I(R4) : | -0.00078125 | device_cu |
| I(R3) : | -0.00046875 | device_cu |
| I(R2) : | 0.00164063 | device_cu |

## $010_{(2)} \rightarrow 2_{(10)}$

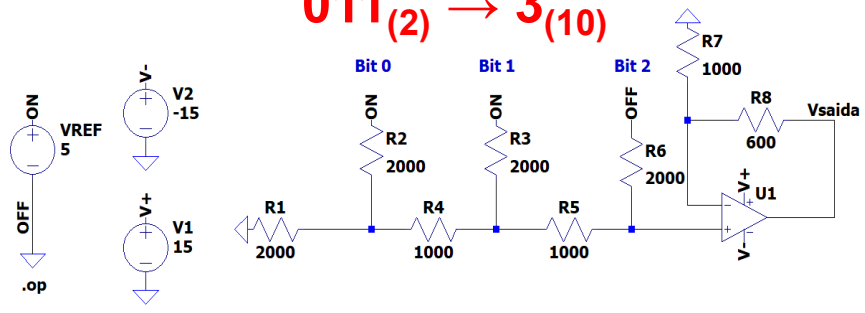



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--- Operating Point ---

| | | |
|---|---|---|
| V(n001) : | 1.25 | voltage |
| V(n004) : | 1.25 | voltage |
| V(vsaida) : | 1.99999 | voltage |
| V(v+) : | 15 | voltage |
| V(v-) : | -15 | voltage |
| V(n002) : | 0.937499 | voltage |
| V(on) : | 5 | voltage |
| V(n003) : | 1.875 | voltage |
| I(R8) : | 0.00125 | device_cu |
| I(R7) : | -0.00125 | device_cu |
| I(R6) : | -0.000624999 | device_cu |
| I(R5) : | -0.000625001 | device_cu |
| I(R4) : | 0.000937499 | device_cu |
| I(R3) : | 0.0015625 | device_cu |

## $011_{(2)} \rightarrow 3_{(10)}$



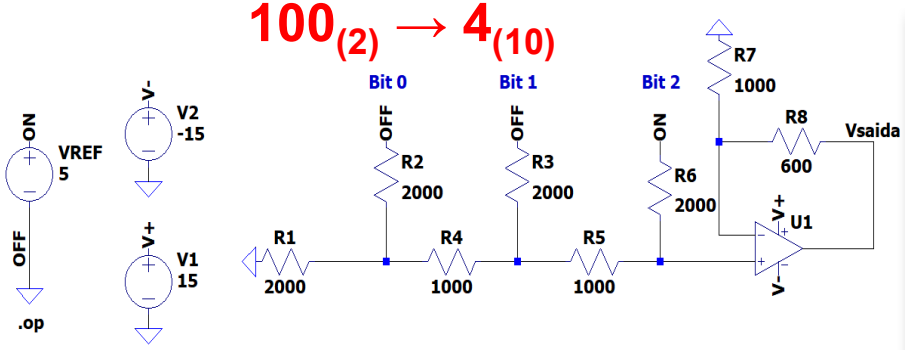

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--- Operating Point ---

| | | |
|---|---|---|
| V(n001) : | 1.87499 | voltage |
| V(n004) : | 1.875 | voltage |
| V(vsaida) : | 2.99999 | voltage |
| V(v+) : | 15 | voltage |
| V(v-) : | -15 | voltage |
| V(n002) : | 2.65625 | voltage |
| V(on) : | 5 | voltage |
| V(n003) : | 2.8125 | voltage |
| I(R8) : | 0.001875 | device_cu |
| I(R7) : | -0.00187499 | device_cu |
| I(R6) : | -0.000937498 | device_cu |
| I(R5) : | -0.000937502 | device_cu |
| I(R4) : | 0.000156249 | device_cu |
| I(R3) : | 0.00109375 | device_cu |

Simulação LTspice

## $100_{(2)} \rightarrow 4_{(10)}$

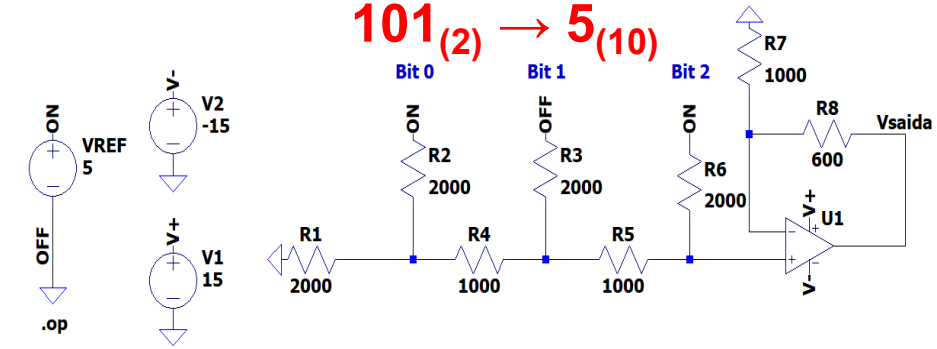



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--- Operating Point ---

| | | |
|---|---|---|
| V(n001) : | 2.49999 | voltage |
| V(n004) : | 2.49999 | voltage |
| V(vsaida) : | 3.99999 | voltage |
| V(v+) : | 15 | voltage |
| V(v-) : | -15 | voltage |
| V(n002) : | 0.624999 | voltage |
| V(on) : | 5 | voltage |
| V(n003) : | 1.25 | voltage |
| I(R8) : | 0.0025 | device_current |
| I(R7) : | -0.00249999 | device_current |
| I(R6) : | 0.00125 | device_current |
| I(R5) : | 0.00125 | device_current |
| I(R4) : | 0.000624999 | device_current |
| I(R3) : | -0.000624999 | device_current |

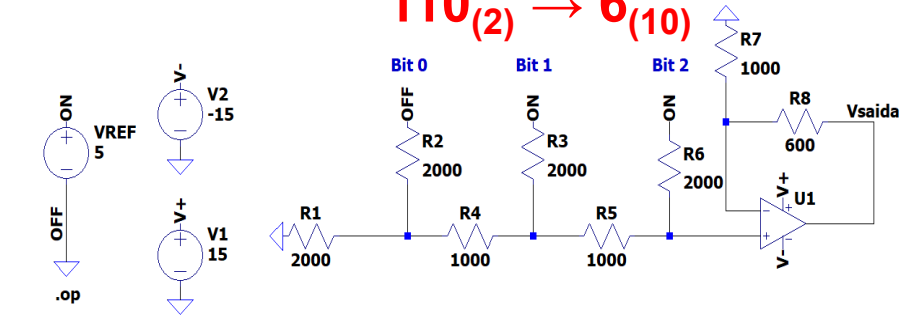
## $101_{(2)} \rightarrow 5_{(10)}$




--- Operating Point ---

| | | |
|---|---|---|
| V(n001) : | 3.12499 | voltage |
| V(n004) : | 3.12499 | voltage |
| V(vsaida) : | 4.99999 | voltage |
| V(v+) : | 15 | voltage |
| V(v-) : | -15 | voltage |
| V(n002) : | 2.34375 | voltage |
| V(on) : | 5 | voltage |
| V(n003) : | 2.1875 | voltage |
| I(R8) : | 0.00312499 | device_cu |
| I(R7) : | -0.00312499 | device_cu |
| I(R6) : | 0.000937503 | device_cu |
| I(R5) : | 0.000937497 | device_cu |
| I(R4) : | -0.000156252 | device_cu |
| I(R3) : | -0.00109375 | device_cu |
| I(R2) : | 0.00132813 | device_cu |
| I(R1) : | 0.00117187 | device_cu |

## $110_{(2)} \rightarrow 6_{(10)}$



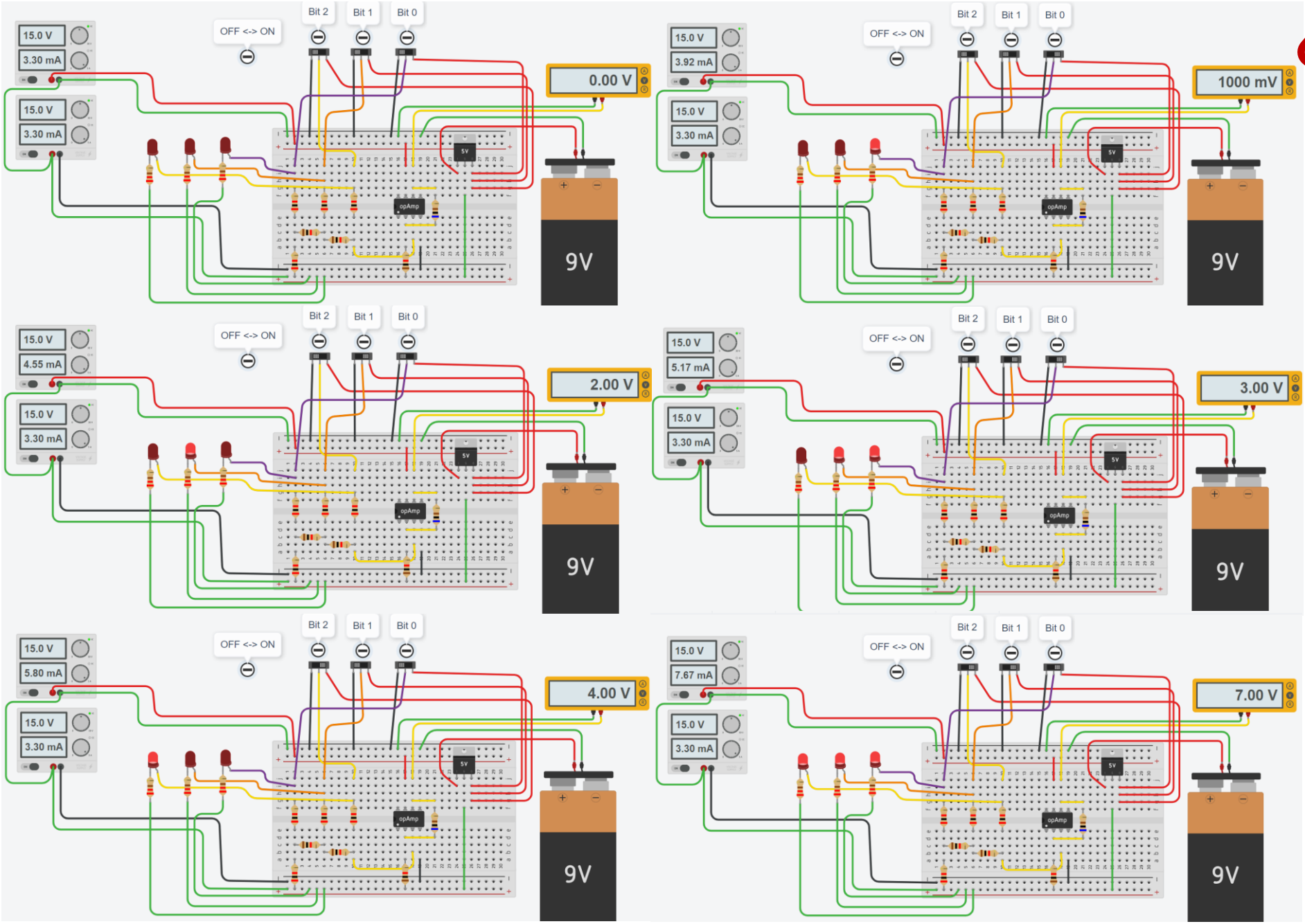

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--- Operating Point ---

| | | |
|---|---|---|
| V(n001) : | 3.74999 | voltage |
| V(n004) : | 3.74999 | voltage |
| V(vsaida) : | 5.99998 | voltage |
| V(v+) : | 15 | voltage |
| V(v-) : | -15 | voltage |
| V(n002) : | 1.5625 | voltage |
| V(on) : | 5 | voltage |
| V(n003) : | 3.125 | voltage |
| I(R8) : | 0.00374999 | device_current |
| I(R7) : | -0.00374999 | device_current |
| I(R6) : | 0.000625004 | device_current |
| I(R5) : | 0.000624996 | device_current |
| I(R4) : | 0.0015625 | device_current |



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# Conversão Digital-Analógica

## CDA → Montagem baseada na escada R – 2R

https://www.tinkercad.com/things/adh4GRzmcJL?sharecode=Lm7A0MREwQeJ_k3GyXO7gCKwxi7A6DO52nPhFtn_iKQ

***Nesta montagem é necessário forçar: OFF → 0V***

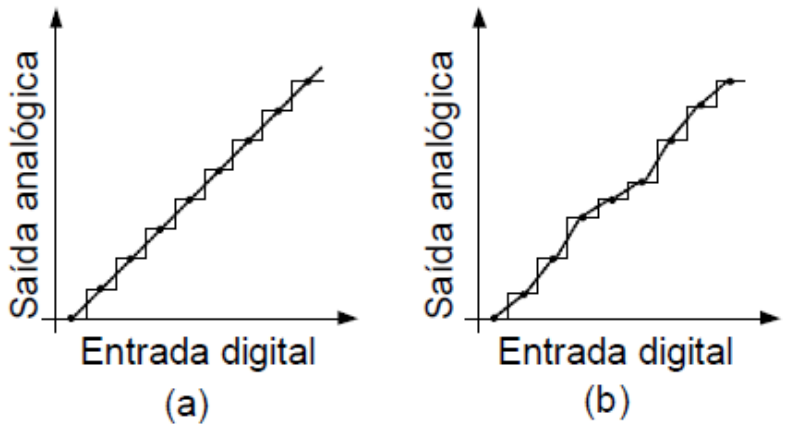

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# Alguns parâmetros de um CDA

$$\text{resolução}\,(\%) = \frac{1}{2^N - 1} \times 100 \quad (7.74)$$

←[1]→

**Figura 7.74. Linearidade: (*a*) CDA ideal e (*b*) CDA real**




←[1]→

**Figura 7.75. Erro: (*a*) Ganho e (*b*) *Offset***

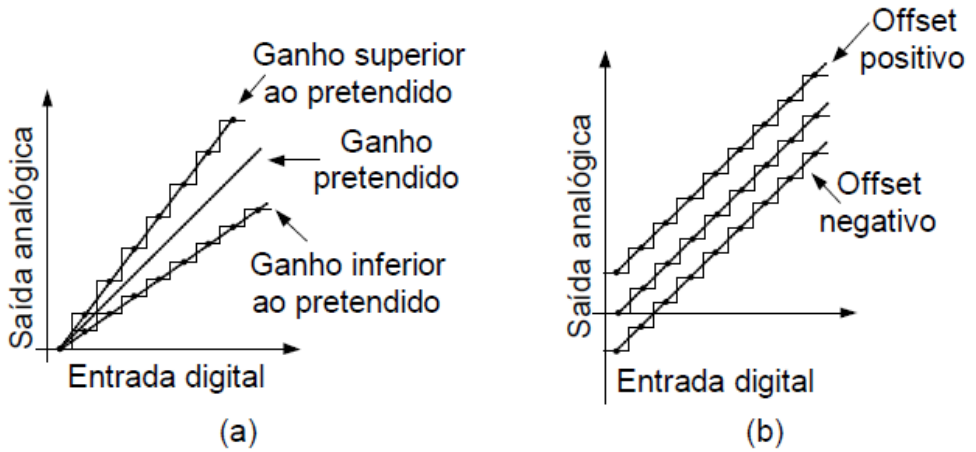
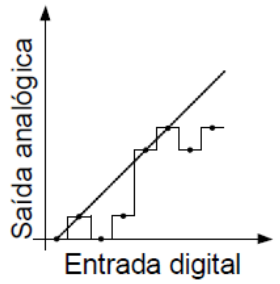



**Figura 7.76. Erro de monotonia para um CAD de 3 *bits***




Sem erro monotonia

| $D_2$ | $D_1$ | $D_0$ | Saída |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 2 |
| 0 | 1 | 1 | 3 |
| 1 | 0 | 0 | 4 |
| 1 | 0 | 1 | 5 |
| 1 | 1 | 0 | 6 |
| 1 | 1 | 1 | 7 |

Com erro monotonia ($D_1$=0)

| $D_2$ | $D_1$ | $D_0$ | Saída |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 4 |
| 1 | 0 | 1 | 5 |
| 1 | 0 | 0 | 4 |
| 1 | 0 | 1 | 5 |



- *Sensibilidade à temperatura*
- $Tempo\ de\ conversão = Tempo\ de\ propagação + Tempo\ de\ estabelecimento$

←[1]


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# Conversão Analógica-Digital

## Sample and hold circuit

**Figura 7.77. Principais elementos que compõe um circuito amostrador-retentor**

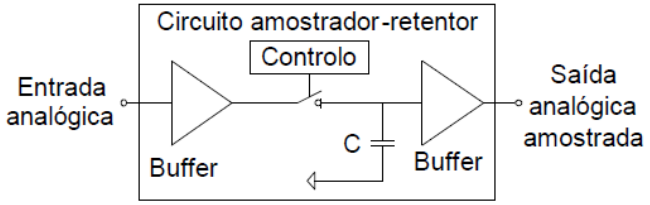
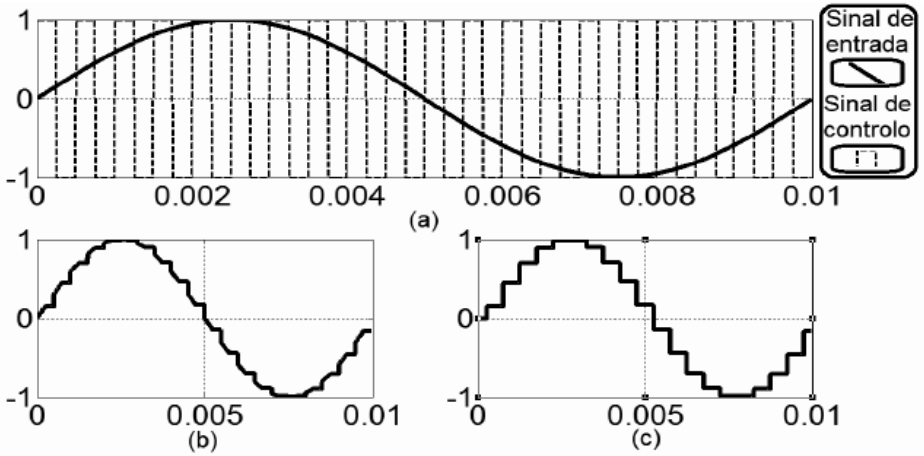


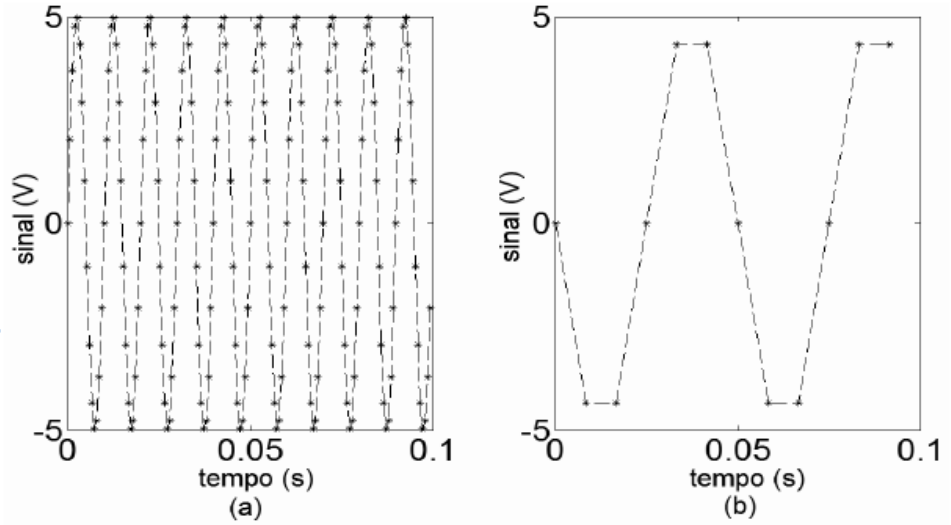

**Figura 7.78. Processo de amostragem e armazenamento: (*a*) sinal analógico original, (*b*) sinal armazenado no condensador e (*c*) sinal discreto final**




## Teorema *Nyquist* versus *aliasing*

**Figura 7.81. Onda sinusoidal com uma frequência de 100 Hz e amplitude 5 Volts: (*a*) reconstrução do sinal original para uma frequência de amostragem de 1 kHz, (*b*) reconstrução do sinal original para uma frequência de amostragem de 120 Hz**

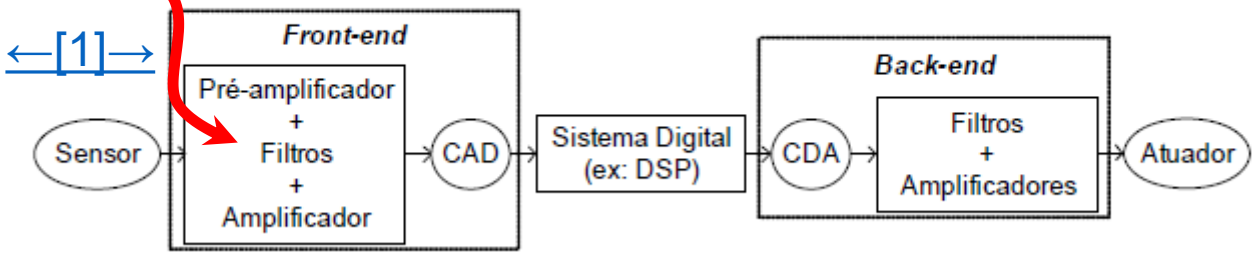
←[1]→




## Filtros anti-aliasing

**Figura 7.68. Diagrama de blocos simplificado de um sistema eletrónico**

←[1]→





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[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

# Conversão Analógica-Digital

## Qualidade do sinal digitalizado:

**Figura 7.83. Diagrama de blocos de um CAD completo**

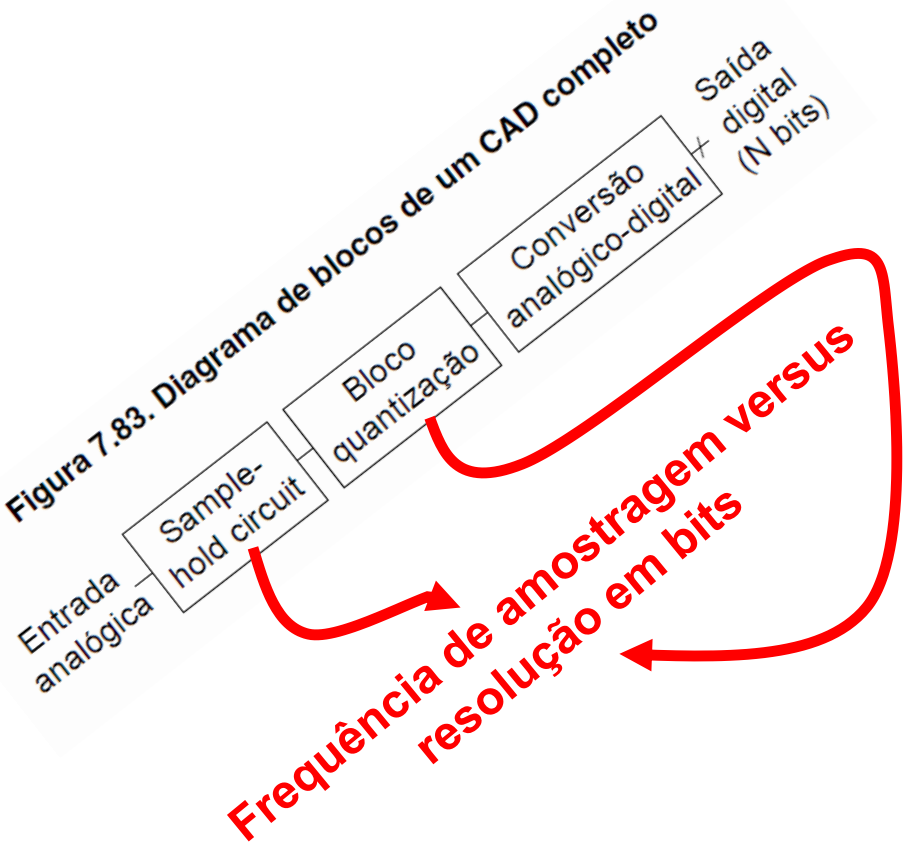
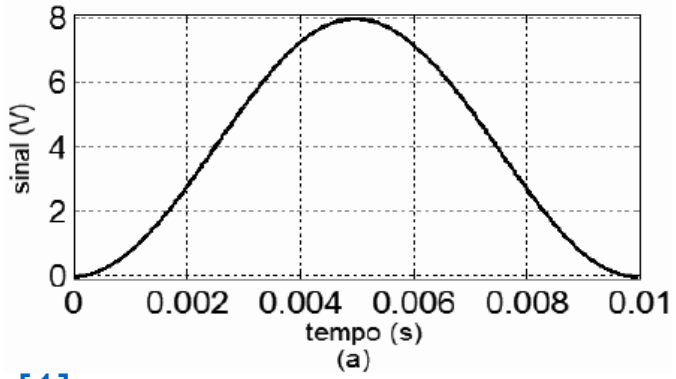
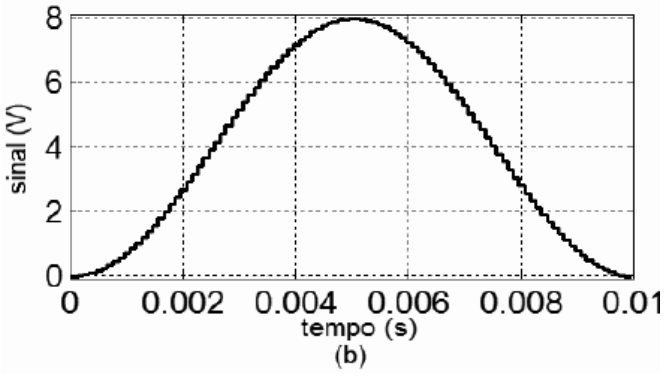



**Figura 7.85. Processo de amostragem (para uma frequência de 10 kHz) e quantização (2 e 4 *bits*):**

(*a*) tensão de entrada do CAD

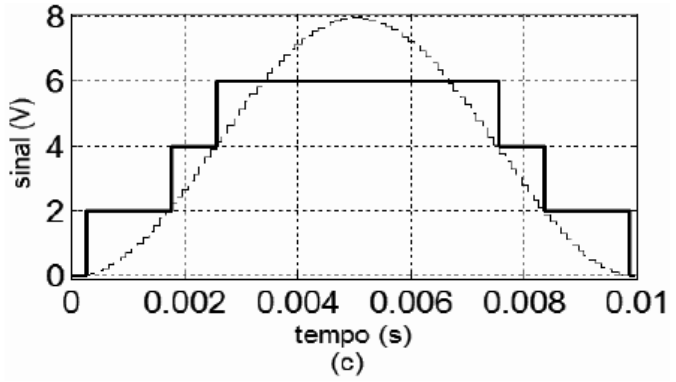



(*b*) tensão de saída do circuito amostrador-retentor

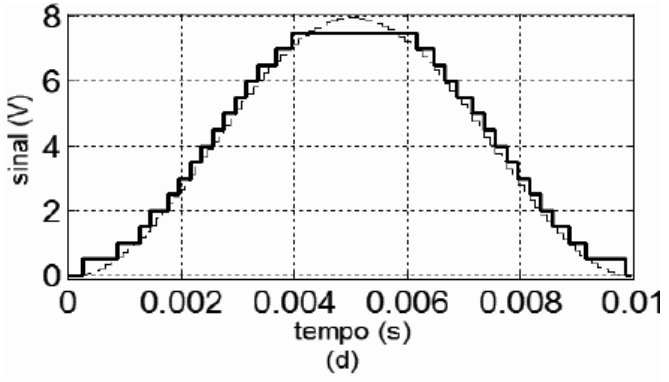



←[1]→

(*c*) quantização do sinal apresentado em (*b*) para uma resolução de 2 *bits*[26]




(*d*) quantização do sinal apresentado em (*b*) para uma resolução de 4 *bits*[26]





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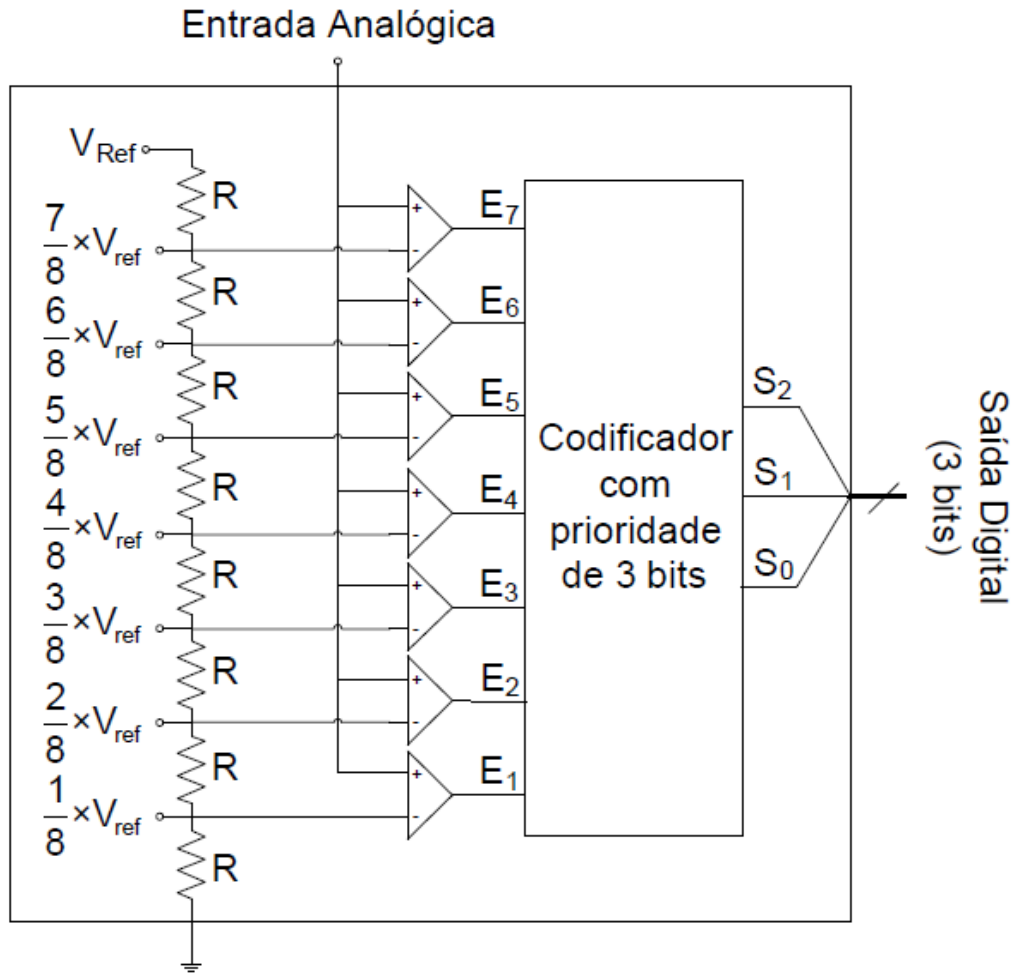
# Conversão Analógica-Digital

## *Parâmetros de um CAD:*

1. ***Linearidade.***
2. ***Erro de Offset.***
3. ***Monótono.***
4. ***Tempo de conversão.***
5. ***Sensível à temperatura.***
6. ***Exatidão***
7. ***Precisão***
8. ***Largura de banda.***

←[1]→

**Figura 7.102. Composição de um CAD do tipo *flash***





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